

Lab 1: FIND-S Algorithm

Aim

To implement the **FIND-S algorithm** and find the most specific hypothesis consistent with the given positive training examples.

Algorithm

1. Initialize the hypothesis with the most specific values.
2. Create the training dataset.
3. Consider only the positive examples.
4. Compare each positive example with the current hypothesis.
5. If an attribute value differs, replace it with ?.
6. Repeat for all positive examples.
7. Display the final hypothesis.

Program

```
# Lab 1: FIND-S Algorithm
```

```
import numpy as np
```

```
# Training data
```

```
data = np.array([
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
])
```

```
# Initialize the most specific hypothesis
```

```
hypothesis = ['0'] * 6
```

```
# Apply FIND-S algorithm
```

```
for row in data:
```

```
    if row[-1] == 'Yes':
```

```
        for i in range(len(hypothesis)):
```

```
            if hypothesis[i] == '0':
```

```
                hypothesis[i] = row[i]
```

```
            elif hypothesis[i] != row[i]:
```

```
                hypothesis[i] = '?'
```

```
print("Most specific hypothesis:", hypothesis)
```

Sample Output

```
Most specific hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

Result

Thus, the **FIND-S algorithm was successfully implemented**, and the most specific hypothesis consistent with the positive training examples was obtained.